

restarts in late winter. Suitable for alpine houses. Propagate by root cuttings in winter or by division in early spring. **W. candida** illus. p.348.

WELWITSCHIA

WELWITSCHIACEAE

Genus of one species of evergreen, desert-growing perennial with a deep tap root. Has only 2 leaves, which lie on the ground and grow continuously from the base for up to 100 years. Frost tender. Needs sun and sharply drained soil. Requires desert conditions: may succeed in a mixture of stone chippings and leaf mold, in a piece of drainpipe to take its long tap root. Propagate by seed when ripe. **W. bainesii**. See *W. mirabilis*. **W. mirabilis**, syn. *W. bainesii*. Evergreen perennial with a short, woody trunk. H to 1ft (30cm), S indefinite. Has 2 strap-shaped leaves, to 8ft (2.5m) long, with tips splitting to form many tendrillike strips. Bears small, reddish-brown cones. Z12 (min. 66°F/19°C)

WESTRINGIA

LAMIACEAE/LABIATAE

Genus of evergreen shrubs, grown for their flowers and overall appearance. Good for a border, or as a hedge or screen, in mild regions. Frost tender. Requires full light and well-drained soil. Water potted specimens moderately, less when not in full growth. Propagate by seed in spring or by semiripe cuttings in late summer. **W. fruticosa**, syn. *W. rosmariniformis* (Australian rosemary), illus. p.453. **W. rosmariniformis**. See *W. fruticosa*.

Wigginsia. See *Parodia*. **Wilcoxia albiflora**. See *Echinocereus leucanthus*. **Wilcoxia schmollii**. See *Echinocereus schmollii*. x **Wilsonara Hambühren Stern gx ‘Cheam’**. See *Oncidium Hambühren Stern gx ‘Cheam’*. **Wintera aromatica**. See *Drimys winteri*.

WISTERIA

PAPILIONACEAE/LEGUMINOSAE

Genus of deciduous, woody-stemmed, twining climbers, grown for their spectacular, fragrant, pealike flowers. Flowers are followed by beanlike, pendant, green, seed pods. Suitable for walls, pergolas, and growing against buildings and trees. Fully hardy. Needs a position in sun and in fertile, well-drained soil. Prune after flowering and again in late winter. Propagate by bench grafting in winter or by seed in fall or spring. Plants grown from seed may not flower for some years and often have poor flowers. ☉All parts may cause severe discomfort if ingested. See also feature panel p.205. **W. brachybotrys** (Silky wisteria). Deciduous, woody-stemmed, twining climber. H to 28ft (9m) or more. Leaves are 8–14in (20–35cm) long, each with 9–13 ovate leaflets. Produces 4–6in (10–15cm) long racemes of scented, pealike, violet to

white flowers, with yellow-marked upper petals, in mid- to late spring, followed by velvety green seed pods. Z5–9. **‘Alba’** see *W.b. f. albiflora* ‘Shiro-kapitan’. **f. albiflora ‘Shiro-kapitan’** (syn. *W.b. ‘Alba’*, *W.b. ‘Shiro-kapitan’*, *W.b. ‘White Silk’*, *W. venusta*, *W.v. ‘Alba’*; illus. p.205) produces very fragrant, occasionally double, white flowers, with a yellow stain at the base of each standard. **‘Murasaki-kapitan’** see *W. x valderi* ‘Murasaki-kapitan’. **‘Okayama’** has violet-blue flower buds, opening paler; highly scented. **‘Shiro-kapitan’** see *W.b. f. albiflora* ‘Shiro-kapitan’. **‘Showa-beni’** has lilac-pink flowers with a greenish yellow mark on each standard. **‘White Silk’** see *W.b. f. albiflora* ‘Shiro-kapitan’. **W. chinensis**. See *W. sinensis*. **W. floribunda** (Japanese wisteria). Vigorous twining climber. H 28ft (9m) or more. Pinnate leaves comprise 11–19 ovate to lance-shaped leaflets. In early summer, pealike, fragrant, blue to violet, pink, or white flowers, the standards marked with white and yellow, are borne in pendent racemes, to 1ft (30cm) or more long, the flowers opening gradually from the bases to the tips; they are often followed by beanlike, velvety green seed pods, to 6in (15cm) long. Z5–9. **‘Alba’** see *W.f. f. alba* ‘Shiro-noda’. **f. alba ‘Shiro-noda’** (syn. *W.f. ‘Alba’*, *W.f. ‘Shiro-noda’*; illus. p.196) bears white flowers in racemes 2ft (60cm) long. **‘Black Dragon’** see *W.f. ‘Kokuryu’*. **‘Domino’** produces blue-violet flowers with light purple-blue standards, marked yellow at the base, and darker purple-blue wing and keel petals, in spring and sometimes again in fall. **‘Double Black Dragon’** see *W.f. f. pleniflora* ‘Yae-kokuryu’. **‘Hon-beni’** see *W.f. f. rosea* ‘Honi-beni’. **‘Kokuryu’** (syn. *W.f. ‘Black Dragon’*) is early- and free-flowering, bearing racemes, 1–1½ft (30–45cm) long, of about 100 scented, dark purplish-violet flowers, each with a pale yellowish-white blotch on the standard petal. **‘Lawrence’** has densely packed racemes of pale violet-blue flowers, with greenish-yellow bases on the standards, and darker blue-violet keel and wing petals, borne in spring. **‘Macrobotrys’** see *W.f. f. multijuga* ‘yushaku’. **‘Multijuga’** see *W.f. f. multijuga* ‘yushaku’. **f. multijuga ‘yushaku’** (syn. *W.f. ‘Macrobotrys’*, *W.f. ‘Multijuga’*) bears lilac-blue flowers in racemes 3–4ft (90–120cm) long. **f. pleniflora ‘Yae-kokuryu’** (syn. *W.f. ‘Double Black Dragon’*, *W.f. ‘Violacea Plena’*, *W.f. ‘Yae-kokuryu’*; illus. p.205) bears racemes 14in (35cm) long, of slightly ragged, fully double, lilac-purple flowers with dark purple-indigo centers. **‘Rosea’** see *W.f. f. rosea* ‘Honi-beni’. **f. rosea ‘Honi-beni’** (syn. *W.f. ‘Honi-beni’*, *W.f. ‘Rosea’*; illus. p.205) has racemes 12–16in (30–40cm) long of dark purplish-pink flowers with a yellow mark on the standard bases. **‘Shiro-noda’** see *W.f. f. alba* ‘Shiro-noda’. **‘Violacea Plena’** see *W.f. f. pleniflora* ‘Yae-kokuryu’. **‘Yae-kokuryu’** see *W.f. f. pleniflora* ‘Yae-kokuryu’. **W. x formosa**. Vigorous twining climber with pinnate leaves comprising 9–15,

broadly ovate to elliptic leaflets. H 28ft (9m) or more. Produces pealike, fragrant, violet-blue flowers, with white and yellow markings, in pendent racemes, to 10in (25cm) long, in late spring and early summer, often followed by beanlike, velvety green seed pods, to 15cm (6in) long. Z5–9 **W. frutescens** (American wisteria; illus p.205). Deciduous, woody-stemmed, twining climber. H 25–30ft (8–10m). Pinnate leaves comprise 9–15 bright green, oblong to lance-shaped leaflets. In early summer, produces fragrant, lilac to pale purple, pealike flowers in pendant clusters. Beanlike, olive-green seed pods, 3–4in (8–10cm) long, form in late summer and persist into winter. Z5–9. **var. frutescens ‘Amethyst Falls’** (illus. p.205) bears flowers with a bluish tint. **var. frutescens ‘Longwood Purple’** (illus. p.205) produces deeper purple blooms. **W. sinensis**, syn. *W. chinensis* (Chinese wisteria). Vigorous, deciduous, woody-stemmed, twining climber. H to 100ft (30m). Has leaves comprising 11 elliptic or ovate leaflets. Produces fragrant, lilac or pale violet flowers, in racemes 8–12in (20–30cm) long, in early summer, followed by velvety seed pods. Z5–9. **‘Alba’** see *W.s. f. alba* ‘Jako’. **f. alba ‘Jako’** (syn. *W.s. ‘Alba’*) has strongly scented, pealike, white flowers in racemes 8–12in (25–30cm) long. **‘Amethyst’** has deep bronze young foliage and bears very fragrant, deep purplish-blue flowers with paler standards. **‘Prolific’** (illus. p.205), H to 100ft (30m), is vigorous, producing abundant, single, lilac-blue to pale violet-blue flowers in longer racemes. **W. x valderi ‘Burford’**. Vigorous, twining, hybrid climber. H 12–25ft (4–8m). In late spring to early summer, bears abundant, dense racemes, to 1½ft (45cm) long, of fragrant, pale violet flowers, with a pale, yellow-white blotch on the standard petal, and darker blue-violet wings and keel. Z5–9. **‘Murasaki-kapitan’** (syn. *W. brachybotrys* ‘Murasaki-kapitan’, *W. venusta* ‘Violacea’) has deep blue-violet flowers with prominent white, slightly yellow-tinged markings on the standards. **W. venusta**. See *W. brachybotrys f. albiflora* ‘Shiro-kapitan’. **‘Alba’** see *W. brachybotrys f. albiflora* ‘Shiro-kapitan’. **‘Violacea’** see *W. x valderi* ‘Murasaki-kapitan’.

WOLFFIA

Duckweed

ARACEAE

Genus of semievergreen, perennial, floating water plants, grown for their curiosity value as the smallest-known flowering plants. Frost hardy to frost tender. Needs a sunny position. Ideal for cold-water aquariums; in warmer climates can be grown in an outdoor pool. Remove excess plantlets as required. Propagate by redistribution of plantlets. **W. arrhiza** (Least duckweed). Semievergreen, perennial, floating water plant. S indefinite. Rootless, rounded, bright green fronds are ¼in (1mm) across. Insignificant, green-tinged, white flowers are produced in summer. Z10–11

WULFENIA

WOLLEMIA

Wollemi pine

ARAUCARIACEAE

Genus of one species of evergreen conifer with narrowly oblong, pointed, needlelike, dark green leaves set opposite each other along shoot. See also CONIFERS. **W. nobilis** (Wollemi pine) illus. p.96.

WOODSIA

WOODSIACEAE

Genus of deciduous ferns, suitable for rock gardens and alpine houses. Fully to frost hardy. Tolerates sun or partial shade. May be difficult to cultivate: soil must provide constant moisture and also be quick-draining, and crowns of plants must sit above soil to avoid rotting. Propagate by division in early spring. **W. obtusa**. Deciduous, tufted, terrestrial or rock-dwelling fern. H 6–12in (15–30cm), S 8in (20cm). In early spring, produces lance-shaped to ovate, pale green fronds, to 1ft (30cm) long; each is composed of 8–15 pairs of elliptic to ovate pinnae, with blunt-tipped segments, covered in somewhat sticky hairs. Z4–8 **W. polystichoides** (Holly-fern woodsia). Deciduous, tufted fern. H 4–12in (10–30cm), S 8–16in (20–40cm). In early spring, has lance-shaped, divided, pale green fronds, to 14in (35cm) long, softly hairy on both surfaces and scaly beneath; each is composed of 15–30 pairs of narrowly sickle-shaped or oblong pinnae, with slightly toothed margins. May be damaged by late frosts. Z4–8

WOODWARDIA

BLECHNACEAE

Genus of evergreen or deciduous ferns, ideal for clothing a moist, shady bank, especially near water. Fully to frost hardy. Prefers partial shade and fibrous, moist, peaty soil. Remove faded fronds regularly. Propagate by division in spring. **W. fimbriata** (Giant chain fern). Evergreen fern. H 3–6ft (1–2m), S 5ft (1.5m). Produces lance-shaped or elliptic, pinnate, mid-green fronds, up to 8ft (2.5m) long. Z9–11 **W. unigemmata** (Asian chain fern). Evergreen fern. H 3ft (1m), S 10ft (3m). Has arching, broadly lance-shaped, pinnate, dark green fronds, to 6ft (2m) long. New foliage emerges brilliant red and fades to brown and then green. (borderline) Z7–10

WULFENIA

PLANTAGINACEAE/SCROPHULARIACEAE

Genus of evergreen, rosette-forming perennials, grown mainly for their spike-like racemes of tubular, 2-lipped flowers. Fully hardy. Suitable for an alpine house. Needs full sun and well-drained soil. Propagate by division in spring or by seed in fall. **W. amherstiana** illus. p.341. **W. carinthiaca**. Evergreen perennial. H and S 10in (25cm). Produces rosettes of oblong to ovate, toothed, dark green leaves, hairy beneath. In summer, bears dense, one-sided racemes of small, tubular, violet-blue flowers on erect stems. Z5–8